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Preserved somatosensory discrimination predicts consciousness recovery in unresponsive wakefulness syndrome

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H I G H L I G H T S

- A P300-based brain computer interface may increase diagnostic precision in patients with disturbances of consciousness (DOC).
- Somatosensory discrimination accuracy predicts the clinical outcome in DOC patients.
- Brain-computer interfaces detect neural correlates of consciousness.

A B S T R A C T

Objective: To assess somatosensory discrimination and command following using a vibrotactile P300-based Brain-Computer Interface (BCI) in Unresponsive Wakefulness Syndrome (UWS), and investigate the predictive role of this cognitive process on the clinical outcomes.
Methods: Thirteen UWS patients and six healthy controls each participated in two experimental runs in which they were instructed to count vibrotactile stimuli delivered to the left or right wrist. A BCI determined each subject's task performance based on EEG measures. All of the patients were followed up six months after the BCI assessment, and correlations analysis between accuracy rates and clinical outcome were investigated.
Results: Four UWS patients demonstrated clear EEG-based indices of task following in one or both paradigms, which did not correlate with clinical factors. The efficacy of somatosensory discrimination strongly correlated (VT2: $R = 0.89$, $p = 0.0000002$, VT3: $R = 0.81$, $p = 0.002$) with the clinical outcome at 6-months. The BCI system also yielded the expected results with healthy controls.
Conclusions: Neurophysiological correlates of somatosensory discrimination can be detected in clinically unresponsive patients and are associated with recovery of behavioural responsiveness at six months.
Significance: Quantitative measurements of somatosensory discrimination may increase the diagnostic accuracy of persons with DOCs and provide useful prognostic information.

1. Introduction

1.1. Disorders of Consciousness (DOCs)

Unresponsive Wakefulness Syndrome (UWS) is a disorder of consciousness (DOC) characterized by spontaneous eye opening in the absence of consistent behavioral responses to external stim-

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uli. When reproducible verbal or motor signs of awareness are detected, the clinical condition is defined as Minimally Conscious State (MCS). In clinical practice, different bedside assessment tools are used to detect purposeful behavior and discriminate between DOCs. Among them, the Coma Recovery Scale-Revised (CRS-R, Kalmar and Giacino, 2005) is considered the most sensitive tool. It explores auditory, visual, verbal and motor responsiveness, as well as communication functions and the level of arousal.

However, the behavioral assessment of awareness has objective limitations, and it has been estimated that up to 30–40% of MCS patients are erroneously classified as UWS (Giacino et al., 2014). This high rate of misdiagnosis amongst the DOCs is particularly impactful due to the ethical debates involving end-of-life issues (Wijdicks, 2008) and affects clinical decision-making involving several aspects of care (e.g. withdrawal of life support, pain treatment, and rehabilitation programs, Lejeune, 2016).

1.2. Neuroimaging and DOCs

Different neuroimaging approaches have been applied to the detection of consciousness, primarily to increase diagnostic accuracy and allow communication. For instance, using fMRI, Monti et al. (2010) showed that motor and visuospatial imagery tasks might reveal covert awareness in unresponsive patients and even allow communication in a single patient with MCS.

More recently, the analysis of brain metabolism by ^{18}F-FDG PET have been used to discriminate among different DOCs by the preservation of brain activity. This methodology was more sensitive than the clinical assessment of the consciousness in the detection of minimal responsiveness and could predict negative outcomes in 92% of the cases (Stender et al., 2014). However, these neuroimaging-based diagnostic approaches have substantial practical limitations, including cost, portability, and viability with patients who rely on metal implants or electronic devices. Hence, they are rarely feasible outside of highly specialized clinical centers.

Several electrophysiological techniques have been applied to the study of the DOCs, with the aim of identifying possible markers of consciousness (Hauger et al., 2016). Unlike some other functional neuroimaging approaches, the EEG-based protocols provide inexpensive assessment tools that can be easily repeated at the patient's bedside and could be extended to several clinical settings.

Brain-computer interfaces (BCIs) are systems that provide a direct connection between brain activity and computer systems. Thus, BCIs can bypass the body's normal output pathways, which may be unavailable due to injury, disease, or other causes. BCI researchers have classically focused on allowing communication and device control, but this has been changing over the last several years (Wolpaw and Wolpaw, 2012). Many recent BCI publications have focused on new ways to help different patient groups, including using BCI technology to assess awareness, provide communication, and support rehabilitation in persons with DOC (e.g., Risetti et al., 2013; Chaudhary et al., 2016; Lesenfants et al., 2016; Real et al., 2016; Guger et al., 2017a; Ortner et al., 2017).

Like other BCIs, BCIs for DOC patients are based on the principle that performing a mental task (such as discriminating among target and non-target stimuli) can elicit distinct EEG signals that can be used to infer conscious awareness. BCIs for DOC patients often (but not always) rely on auditory or vibrotactile stimuli, since DOC patients typically cannot use visual-based interfaces. For instance, the presentation of the patient's own photo amidst unfamiliar photos could elicit responsiveness in UWS and MCS patients (Pan et al., 2014). Similarly, the subject's own name (SON) paradigm, auditory oddball paradigms involving discrimination of a target word or tone, and other auditory paradigms could reveal passive recognition and/or active command following in UWS and/or MCS patients unable to express behavioral responses (Lulè et al., 2013; Risetti et al., 2013; Ortner et al., 2017). BCIs based on vibrotactile P300 evoked potentials have also been validated for detection of consciousness and communication in healthy subjects and LIS/CLIS patients (Ortner et al., 2014; Spataro et al., 2016; Guger et al., 2017b). This approach is depicted in Fig. 1.

1.3. Outcome prediction

There are no standardized clinical measures for predicting DOC patient outcomes. Since outcome prediction depends largely on the neurologist's assessment of each patient's remaining function, and conventional scales may produce inaccurate assessments, improved assessment tools could facilitate outcome prediction. Beyond facilitating assessment through improved diagnostic accuracy, EEG activity and BCI performance might also directly predict patient outcomes.

Adapting EEG-based BCI technology to the prediction of clinical outcomes is even newer, and has shown some promising results (Juan et al., 2016; Tzovara et al., 2016; Wang et al., 2017; Zubler et al., 2017). This new direction could help doctors, patients, and families consider and plan for likely clinical outcomes, and might influence decisions regarding medication, treatment, rehabilitation, and other issues.

In the present study, we used the same vibrotactile BCI system used in our recent prior work to explore somatosensory attention in UWS patients. Our goals were to: (1) further validate the efficacy of this system in the assessment of conscious function and (2) explore the predictive role of these neurophysiological assessments in positive clinical outcomes.

2. Methods and materials

2.1. Participants

We recruited 13 patients in UWS, all resulting from severe brain injury [11 male, 2 females, mean 59 (range 19–91)]. The participants were admitted at three clinical centers (the Emergency Unit at the University Hospital of Palermo, the Rehabilitation Unit at ICSS Salvatore Maugeri in Sciacca, and the Respiratory Intensive Care Unit at ARNAS Hospital in Palermo, all in Italy) between April and June 2016. The median disease duration was 60 days since brain injury. Behavioral responsiveness was repeatedly assessed by trained neurologists using the Coma Recovery Scale-Revised (CRS-R). Six healthy controls were enrolled as a control group and were matched with the patient group in gender and age. Table 1 shows the clinical and demographic characteristics of the participants. Written informed consent was obtained from the healthy controls or the legal guardians of the patients. All procedures were approved by the Ethics Committee Palermo 1.

2.2. Assessment

The experiments were performed using a complete hardware and software platform for stimulus presentation, data recording, and real-time analysis (mindBEAGLE®, g.tec Guger Technologies OG, Austria). It includes a laptop with installed software, three vibrotactile stimulators, two in-ear headphones, one g.USBamp biosignal amplifier with 16 channels and 24 Bit ADC resolution and one EEG cap with 16 g.LADYbird active electrodes. EEG data were recorded from an electrode cap with eight channels (Fz, C3, Cz, C4, CP1, CPz, CP2 and Pz) based on the International 10–20 electrode system, then amplified and sampled at 256 Hz.

All patients participated in a BCI session with two different paradigms (VT2 and VT3). In the VT2 paradigm, one vibrotactile stimulator was placed on each wrist, which produced non-target

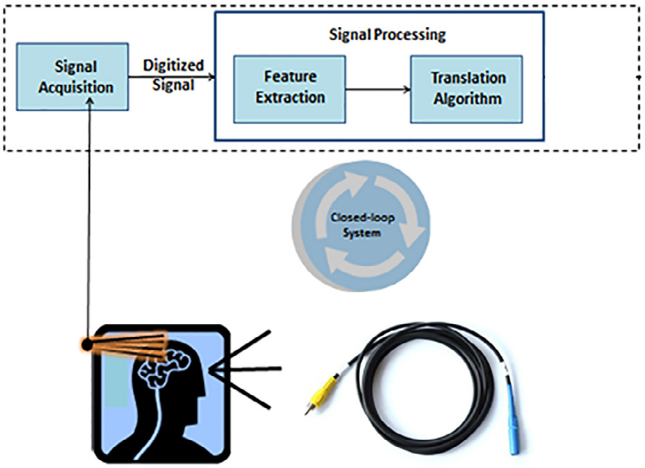

Fig. 1. This schematic presents the approach behind vibrotactile BCIs for persons with DOC. The picture on the bottom right shows one of the vibrotactile stimulators used in this study.

(frequent) and target (rare) stimuli, presented at a ratio of 7:1. All stimuli in both runs lasted 100 ms. In this run, lasting 2.5 min, the subjects were verbally instructed to mentally count the target stimuli.

In the VT3 paradigm, the left and right vibrotactile stimulators are equiprobable (12.5%), and a third stimulator placed on the back receives 75% of the stimuli, acting as a distractor. Thus, each sequence of eight stimuli was presented at a ratio of 1:1:6. Each VT3 run lasted 3 min and contained four trial blocks. At the beginning of each trial block, the subject was verbally instructed by headphones to mentally count stimuli on the right or the left wrist. Across these four trial blocks, the left and right wrists were each designated as targets twice in pseudorandom order. Each trial block presented 30 sequences of eight stimuli, and each group of eight stimuli included exactly one target stimulus. As with the VT2 paradigm, the order of stimulus presentation was determined pseudorandomly within each group of eight stimuli. Fig. 2 further illustrates the timing of the experimental paradigm.

All patients were followed up until death or 6 months after ictus in survivors. The principal clinical outcome measure was CRS-R, which was evaluated by blinded assessors.

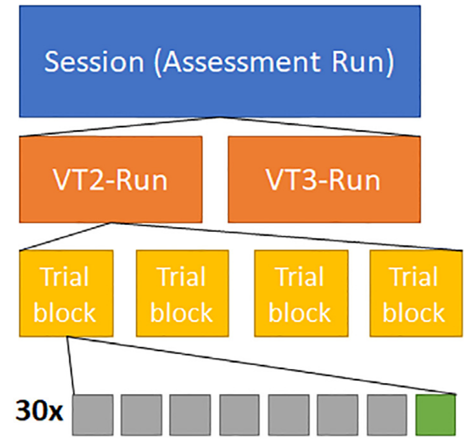

Fig. 2. This figure presents the experimental paradigm. Each session had one VT2 run and one VT3 run. Each run contained four trial blocks. Each trial block had 30 groups of eight stimuli.

2.3. Signal processing and classification

The raw EEG data and the stimulation time points were recorded during each run. Data epochs from −100 ms to 600 ms from each stimulation point were created and baseline corrected. The data were notch-filtered at 50 Hz and bandpass-filtered within 0.1–30 Hz. Trials with an amplitude above 100 μV were automatically rejected.

The data classification used a linear discriminant analysis (LDA) approach (Ortner et al., 2016; Allison et al., 2017). For the cross-validation approach, all 480 trials from each run were split into a training set and a testing set, with 50% of the data in each set. The training set was used to train the LDA classifier, which was then tested on the testing data. The LDA score was calculated for each trial group (7 non-target trials, 1 target trial), and the trial with the highest score was identified as the target. If the classified trial was the real target, the classification was considered correct.

The classification accuracy in the VT2 and VT3 paradigms was based on the percentage of target stimuli classified correctly divided by the total number of target stimuli. Since each trial group has 30 groups of eight stimuli, classification accuracy was recalculated after each trial group, providing an estimate of accuracy

Table 1
Characteristics of the participants.

Patient	Recruiting center	Gender	Age	Diagnosis	Disease duration (days)	CRS-R at examination
UWS 1	ICSM	M	19	TBI	360	6
UWS 2	ICSM	M	19	TBI	300	6
UWS 3	ICSM	F	36	TBI	60	3
UWS 4	ICSM	M	34	HBI	240	6
UWS 5	ARNAS	M	91	STROKE	49	6
UWS 6	ARNAS	M	82	SDH	64	5
UWS 7	ARNAS	M	61	HBI	45	6
UWS 8	ARNAS	M	66	STROKE	180	5
UWS 9	DIBIMED	F	65	ME	35	6
UWS 10	DIBIMED	M	60	HBI	45	4
UWS 11	DIBIMED	M	85	SDH	60	5
UWS 12	DIBIMED	M	72	HBI	40	5
UWS 13	DIBIMED	M	78	STROKE	36	6
HC 1	ARNAS	M	68	–	–	–
HC 2	ICSM	F	37	–	–	–
HC 3	ICSM	M	58	–	–	–
HC 4	DIBIMED	F	42	–	–	–
HC 5	DIBIMED	M	43	–	–	–
HC 6	ARNAS	M	38	–	–	–

UWS: Unresponsive Wakefulness Syndrome; ICSM: Istituti Clinici Scientifici Maugeri Sciacca; ARNAS: Respiratory Intensive Care Unit, ARNAS Civico General Hospital, Palermo DIBIMED: Department of Biopathology and Medical Biotechnologies (DIBIMED), University of Palermo, TBI: Traumatic Brain Injury; HBI: Hypoxia-Ischemia Brain Injury; SDH: Subdural Hematoma; ME: meningoencefalitis; HC: Healthy control based on data from 1 through 30 trial groups (Fig. 2). This whole classification procedure is repeated ten times.

The classifier was blind to the correct target and did not group together different stimuli of the same type (such as all non-target or distractor stimuli). Thus, the chance accuracy rate for both VT2 and VT3 paradigms was 12.5%. Recent work to validate the efficacy of the mindBEAGLE system showed that the classifier indeed yields chance accuracy in "sham" versions of the VT2 and VT3 paradigms while yielding high accuracy in the same participants during regular operation (Allison et al., 2017; Guger et al., 2017b).

During and after stimulus presentation, the mindBEAGLE system presents information about the patient's brain activity and the BCI classification accuracy to a system operator. This includes a graph that shows the ERPs elicited by target stimuli and an average of all non-target stimuli, which are grouped together for this display. Time points that are significantly different ($p < 0.05$ according to a Kruskal-Wallis-Test between the target and non-target ERPs) are shaded green.

2.4. Outcome prediction and statistics

Patients who crossed a threshold of responsiveness (VT2/VT3 accuracy rate equal to or greater than 50%) were compared with non-responsive patients. All analyses were made using SIGMASTAT 3.5 and SIGMAPLOT 8.0 (Systat Software Inc., San José, CA, USA) software packages. Continuous variables were compared using the Mann–Whitney *U* test. The Spearman rank correlation coefficient was used to evaluate whether the neural correlates of somatosensory discrimination were associated with the behavioral assessment of consciousness, expressed as CRS-R scores, at six months following the investigation.

3. Results

Table 2 shows the VT2 and VT3 accuracy rates, clinical outcome and CRS-R at six months follow-up. Overall, the patients obtained higher accuracy rates in VT2 than VT3 ($p = 0.04$, paired *t*-test) and performed significantly below the control group in both paradigms (VT2: VT3: $p = 0.001$, paired *t*-test).

In the VT2 paradigm, the average online accuracy of the patients was 31.53% ± 22.58, while the online accuracy of the controls was 92.5% ± 14.05 ($p = 0.000$, paired *t*-test). In the VT3 paradigm, the average online accuracy of the patients was 21.15% ± 26.46, while the online accuracy of the controls was 83.33% ± 13.66 ($p = 0.000$, paired *t*-test). Fig. 3 presents an example of ERPs from one patient in both paradigms, along with classification accuracy based on averages of between 1 and 30 trial groups.

All controls achieved an accuracy greater than the significance level of 23% (well above the 95% confidence interval with a binomial test that gives 23%). Among the patients, patients 1, 7 and 9 showed responsiveness in both VT2 and VT3 paradigms, while patient 8 did so in VT2 only. Patient 10 showed in 46% of the trials artifacts, which can explain the low accuracy. All other subjects had between 0% and 11% trials showing artifacts.

Across the two paradigms, responsive patients did not differ from unresponsive patients in age (VT2: $p = 0.39$; VT3: $p = 0.28$; Mann-Whitney *U* Test), disease duration (VT2: $p = 0.9$, VT3: $p = 0.6$; Mann-Whitney *U* Test) and CRS-R (VT2: $p = 0.33$, VT3: $p = 0.16$; Mann-Whitney *U* Test). Furthermore, the VT2 and VT3 paradigms did not correlate with the CRS-R at the basal evaluation (VT2 *vs* CRS-R, $r = 0.32$, $p = 0.27$; VT3 *vs* CRS-R, $r = 0.32$, $p = 0.26$; Spearman's correlation, Fig. 4A and B).

At 6-months follow-up, patients 2 and 13 passed away because of sepsis, after developing hospital-acquired antibiotic resistant bacterial infections. Patient 5, who was 91 years old, became deceased as a consequence of cardiac arrest. All the responsive patients except patient 8 recovered reproducible responses to commands or oriented voluntary movements. Patients 1 and 7 evolved to MCS, whereas patient 9 recovered consciousness.

Correlation analysis showed a strong association between the VT2 and VT3 accuracy rates and the 6-months CRS-R scores (VT2 *vs* CRS-R, $r = 0.89$, $p = 0.0000002$; VT3 vs CRS-R, $r = 0.81$, $p = 0.002$, Fig. 4C and D). Moreover, the CRS-R score did not correlate with age ($r = 0.23$, $p = 0.5$) or disease duration ($r = 0.1$, $p = 0.7$).

We conclude that VT2 and VT3 accuracy rates at baseline can predict a change of CRS-R, as a marker of clinical recovery. On its

Table 2
Accuracy Rates in % for the VT2 and VT3 paradigms, and clinical outcomes at 6 months follow-up. Bold accuracies show patients that improved. Most patient's data showed no artifacts at all (UWS 1, 2, 4, 8 and 12), some patients had trials with below 11% artifacts (UWS 5, 6, 7, 9, 11, 13) and one patient showed 46% artifacts (UWS 10). All healthy controls had no artifacts at all.

Patient	VT2 accuracy	VT3 accuracy	6 months outcome	6 months CRS-R
UWS 1	**80**	**80**	MCS	15
UWS 2	30	10	Deceased	–
UWS 3	45	10	UWS	6
UWS 4	20	30	UWS	3
UWS 5	30	0	Deceased	–
UWS 6	20	0	UWS	5
UWS 7	**50**	**60**	MCS	13
UWS 8	50	20	UWS	6
UWS 9	**50**	**50**	Awake	23
UWS 10	0	15	UWS	4
UWS 11	20	0	UWS	5
UWS 12	10	0	UWS	5
UWS 13	5	0	Deceased	–
Average	31.53	21.15		
HC 1	90	80	–	–
HC 2	100	90	–	–
HC 3	100	80	–	–
HC 4	100	90	–	–
HC 5	100	60	–	–
HC 6	65	100	–	–
Average	92.5	83.33		

UWS: Unresponsive Wakefulness Syndrome; MCS: Minimally Conscious State; HC: Healthy control.

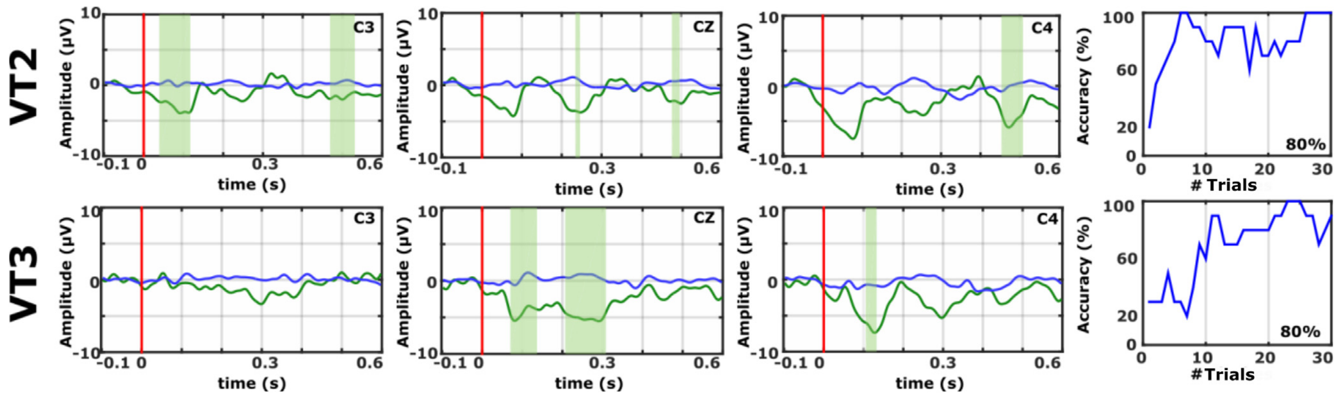

Fig. 3. Evoked Potentials (EP) and accuracy rates for VT2 and VT3 runs from patient UWS1 over electrode positions C3, CZ, and C4. The EP plots contain target EPs in green and non-target EPs in blue. The green shaded areas indicate significant differences between targets and non-targets (Kruskal-Wallis test, $p < 0.05$). The vertical magenta line shows the onset of the vibrotactile stimulus. The P300 complex, particularly in VT3, would appear atypical for a 19-year-old healthy person. However, this negative deflection is not unusual with patients (Guger et al., 2017b).

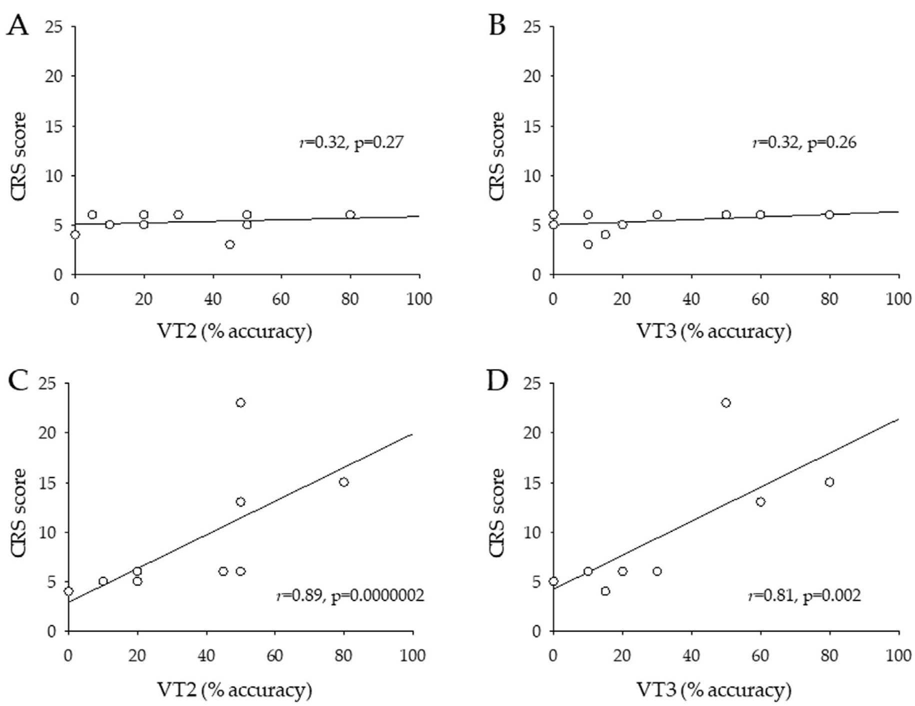

Fig. 4. Correlation between the two vibrotactile stimulator paradigms (VT2 and VT3, evaluated at baseline) and the CRS-R score in UWS patients at baseline (A and B) and after six months (C and D). Analysis was carried out with the Spearman's correlation test. Both VT2 and VT3 results were correlated with the CRS-R improvement at six months.

turn, the change of CRS-R score is not predicted by demographical (*i.e.*, age) or clinical (*i.e.*, disease duration) variables.

4. Discussion

The BCI paradigms used in this pilot study showed brain activity indicating purposeful behavior in one-third of the patients clinically classified as UWS. This result overlaps previous findings involving cognitive event-related potentials and BCIs (Vanhaudenhuyse et al., 2008; for review, see Chaudhary et al., 2016; Lesenfants et al., 2016; Guger et al., 2017b). Furthermore, this result reinforces the clinical utility of integrating behavioral and neurophysiological approaches to increase the diagnostic and predictive accuracy relating to persons with DOC.

To discriminate between voluntary and automatic cognitive processes, we evaluated two different paradigms. The VT2 paradigm explored detection of an infrequent target and right/left somatosensory discrimination, and the VT3 paradigm also evaluated switching tasks (between counting targets to the left or right hand) and suppression of irrelevant stimuli (the distractors). The additional tasks in the VT3 paradigm should reflect additional cognitive processes and influence P300 subcomponents (Polich, 2004). The different complexity of the two tasks resulted in a higher rate of positive responses to the easier task. However, albeit with different sensitivity, both paradigms could predict the recovery of behaviorally detectable responsiveness even several months after the disease's onset. Future research could use different analysis methods to gain more information from the P300 complex, which could lead to improved diagnosis and prediction. Furthermore, analyses of other ERP components, such as the MMN or LPC, and/or additional EEG analyses such as free-running EEG activity, mutual information, connectivity, or evaluation of local probability effects (Fazel-Rezai et al., 2012; Erlbeck et al., 2017; Zubler et al., 2017) may improve diagnoses, prediction, and perhaps other applications. The absence of these additional analyses is a weakness of the present paper, and we are working on a revised version of mindBEAGLE that can conduct additional real-time analyses and support later data analysis.

The average VT2 accuracy for patients was 31.53% and for VT3 21.15%. In comparison, 12 LIS/CLIS patients achieved a mean accuracy of 76.6% in VT2 and 63.1% in VT3 (Guger et al., 2017b), which is better than in UWS patients. The healthy controls in this study achieved 92.5% in VT2 and 83.33% in VT3. UWS 10 showed in 46% artifacts which explains also the low classification accuracy.

The healthy controls' results in this paper further showed that the system and methods used here could correctly identify EEG-based indices of somatosensory attention in healthy persons. This supposition is also supported by our current outcome prediction results with UWS patients and recent results with "sham" operation (Allison et al., 2017; Guger et al., 2017b).

This evidence supports the hypothesis that the preservation of the somatosensory discrimination in comatose states is associated with favorable outcome. We included UWS patients with different etiologies admitted in non-specialized centers, showing that the methodologies used here are feasible in real-world settings. However, this approach still requires considerable further research and validation before widespread clinical adoption. Longitudinal studies on larger samples over longer durations would further detail the sensibility and specificity of different quantitative EEG measures as non-invasive tools for assessment and prediction. New and improved signal processing methods, electrode systems, user interaction protocols, passive and active paradigms, incorporation of other signals or imaging methods, and additional analyses of factors that correlate with outcome are all important future directions. Some of these potential improvements could extend the "hybrid BCI" concept in this system by supplementing its existing tools with other paradigms, signals or imaging methods. Hybridization is a particularly promising direction that many different groups will probably explore shortly, as it could greatly improve the detail and specificity of clinical assessments while providing much greater flexibility and usability to patients (Pfurtscheller et al., 2010; Allison et al., 2012).

Another future direction involves the optimization of the number of trials (or events) that are averaged together prior to classification. Reducing the number of trials could lead to more rapid assessment or communication, but could also impair accuracy. We used a very conservative setting of 30 classifications per average, which is much larger than conventional or even older P300 BCIs (Farwell and Donchin, 1988; Fazel-Rezai et al., 2012; Wolpaw and Wolpaw, 2012). As Fig. 1 indicates, patient UWS1 may have attained adequate accuracy with averages of only ten trials. The ideal number of trials should be decided for each patient based on BCI performance and other factors, ideally including the patient's preferences when possible. Changing the number of trials could also influence the threshold for successful assessment (Yuan et al., 2013).

Similarly, the number of recording sessions and runs required for each patient before reaching conclusions or taking action also needs further research. DOC patients are sometimes unresponsive, or may not produce EEG signals that reflect conscious awareness for other reasons during one or more sessions. Thus, if a patient's EEG does not reflect conscious awareness, possible communication, or likely recovery across one or even a few recording sessions, additional sessions might produce a different result.

Overall, this evidence further demonstrates that EEG-based BCI systems can provide a more detailed assessment of each patient's remaining functions than conventional scales. More importantly, they also support emerging work for prediction of outcome. We have adapted an approach based on somatosensory attention and shown that it appears promising for prediction of outcome after six months. This is an early report from a nascent research direction. The need for additional data analyses, participants, and research into numerous future issues here is a limitation of the current study as well as a challenge for the field.

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Disclosure

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